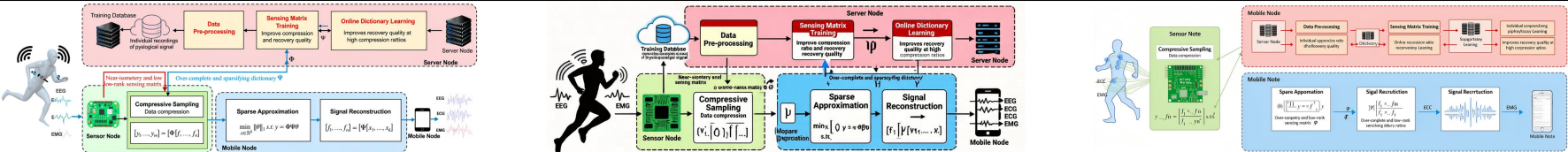
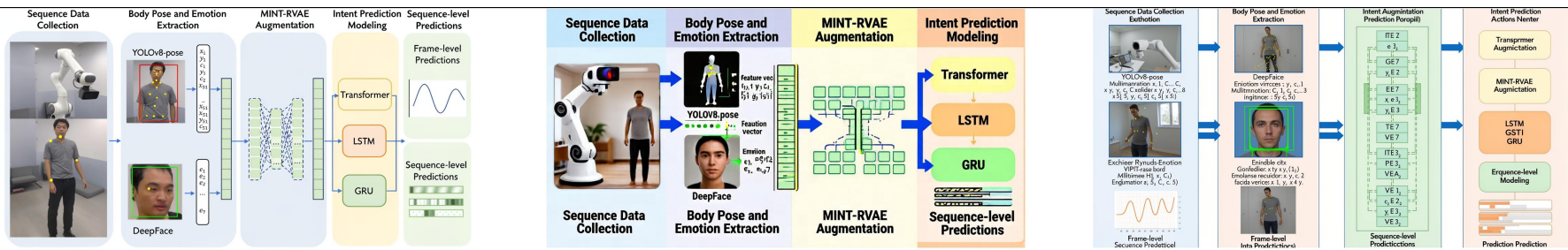


The figure presents a block diagram of a data-driven compressive sensing framework for physiological signal processing, structured into three main computational nodes: Sensor Node, Mobile Node, and Server Node, each represented by distinct background colors—light green, light blue, and light pink respectively—and separated by dashed borders. The overall layout is horizontal, depicting a pipeline from raw physiological data acquisition on the left to reconstructed signals on the right. On the far left, a human silhouette running with wireless signals emanating from body sensors represents the source of physiological signals such as EEG, ECG, and EMG, illustrated with waveform icons. These signals feed into the Sensor Node, which contains a green circuit board icon labeled 'Sensor Node'. Within this node, the first processing step is 'Compressive Sampling' (a white rectangular box with black text), described as 'Data compression'. Below it, a mathematical equation is shown:  $[y_1, \dots, y_n] = [\Phi] [f_1, \dots, f_n]$ , representing the compressive sampling process using a sensing matrix  $\Phi$ . An arrow from the Sensor Node points to this module, labeled 'Near-isometry and low-rank sensing matrix  $\Phi$ ', indicating that  $\Phi$  is trained offline. The compressed data then flows to the Mobile Node (light blue background), which contains two sequential modules: 'Sparse Approximation' and 'Signal Reconstruction'. The Sparse Approximation module includes the optimization problem:  $\min_{\tilde{f}} \|\tilde{f}\|_1, \text{ s.t. } y = \Phi \tilde{f}$ , indicating sparse coding over a dictionary  $\Psi$ . This module receives the dictionary  $\Psi$  from the Server Node via a blue arrow labeled 'Over-complete and sparsifying dictionary  $\Psi$ '. The Signal Reconstruction module follows, with the equation  $[f_1, \dots, f_n] = [\Psi] [y_1, \dots, y_n]$ , showing reconstruction using the learned dictionary. The Mobile Node also outputs reconstructed EEG, ECG, and EMG waveforms. The Server Node (light pink background) at the top contains three modules: 'Data Pre-processing', 'Sensing Matrix Training', and 'Online Dictionary Learning', all in beige boxes with red headers. Data from the 'Training Database' (represented by a cloud and database icon with text 'Individual recordings of physiological signal') enters the 'Data Pre-processing' module. This feeds into 'Sensing Matrix Training', which aims to 'Improve compression ratio and recovery quality'. The output of this module is the sensing matrix  $\Phi$ , which is then used in 'Online Dictionary Learning' to 'Improve recovery quality at high compression ratios'. The Server Node also includes a 'Training Database' and a 'Sensing Matrix' icon. Arrows indicate the flow: Training Database → Data Pre-processing → Sensing Matrix Training → Online Dictionary Learning → Sensing Matrix  $\Phi$  → Mobile Node. A bidirectional arrow labeled ' $\Psi$ ' connects the Online Dictionary Learning module to the Sensing Matrix Training module. The Server Node also includes an icon of stacked servers labeled 'Server Node'. Arrows indicate the flow from Training Database → Data Pre-processing → Sensing Matrix Training → Compressive Sampling → Sparse Approximation → Signal Reconstruction → Mobile Node. A distributed architecture where training occurs centrally on the server, while compression and reconstruction happen at edge devices (sensor and mobile nodes).



The figure illustrates a comprehensive data processing pipeline for human-robot interaction (HRI) intent prediction, structured in a left-to-right flow. The global layout consists of five main stages: Sequence Data Collection, Body Pose and Emotion Extraction, MINT-RVAE Augmentation, Intent Prediction Modeling, and final prediction outputs (Frame-level and Sequence-level Predictions). Each stage is visually separated by distinct background colors and labeled with descriptive captions. In the first stage, 'Sequence Data Collection', a white robotic arm equipped with an RGB camera captures video sequences of a human subject in an indoor environment. The human is shown wearing a gray shirt and dark pants, with yellow dots marking key body joints, indicating pose tracking. This stage feeds into the next phase, 'Body Pose and Emotion Extraction', which processes the captured frames. Two parallel pathways are shown: one extracts body pose using a YOLOv8-pose model, represented by a bounding box around the person with highlighted keypoints, producing a feature vector containing x, y coordinates and confidence scores ( $x_1, y_1, c_1, \dots, x_n, y_n, c_n$ ). The other pathway extracts facial emotion using DeepFace, depicted by a close-up of the face with a green bounding box, generating an emotion vector ( $e_1, e_2, \dots, e_r$ ). These two vectors are then concatenated into a single multimodal feature vector, displayed as a tall vertical rectangle with alternating entries from both sources. The third stage, 'MINT-RVAE Augmentation', takes the concatenated feature vector as input. It is visualized as a variational autoencoder structure with a bottleneck, consisting of multiple layers of light green rectangular nodes connected by dashed lines, representing the encoder-decoder architecture of the MINT-RVAE model. This module is designed to generate synthetic sequences to address class imbalance in HRI datasets. The fourth stage, 'Intent Prediction Modeling', receives the augmented features and routes them through three different neural network architectures: Transformer (light yellow rounded rectangle), LSTM (light orange rounded rectangle), and GRU (light green rounded rectangle). These models are arranged vertically, indicating they are alternative backbones for intent prediction. The final output stage splits into two components: 'Frame-level Predictions' and 'Sequence-level Predictions'. The final output predictions are depicted as three horizontal bars with varying patterns, indicating aggregated or long-term intent classification over time. All connections between modules are indicated by solid blue arrows, showing the unidirectional data flow from data collection to final predictions.



The figure illustrates the architecture of a Target-oriented Grasp Network (TOGNet), designed to enable robots to perform object grasping under multimodal human guidance. The global layout is structured as a left-to-right dataflow pipeline, starting from input RGB-D images on the far left, progressing through multimodal guidance modules at the top center, converging into TOGNet, and culminating in robotic execution on the right. The bottom row shows real-world robotic setup images corresponding to the input and output stages. At the top-left, 'Input RGB-D images' are shown as a rectangular image patch containing a yellow bottle, white spray bottle, small white cup, and orange cup on a black surface, with a blue bar beneath indicating depth data. This input feeds into three parallel multimodal guidance modules: 'Language Instructions', 'Pointing Gesture', and 'Interactive Click'. The 'Language Instructions' module, highlighted in light blue, features a speech bubble with the text 'Can you give me the yellow bottle?' directed toward a user icon. An arrow points from this instruction to a 'Grounded SAM' module, which processes the query and outputs a bounding box (pink rectangle) around the yellow bottle in a cropped image, demonstrating semantic grounding. The 'Pointing Gesture' module, in light beige, shows a hand pointing toward the yellow bottle with a green arrow and a blue dot indicating the target point, showing how spatial gestures are interpreted. The 'Interactive Click' module, in light beige, shows a mouse cursor clicking on the yellow bottle, showing a mouse cursor clicking on the yellow bottle. All three guidance modalities converge via arrows into the central 'Target-Oriented Grasp Network (TOGNet)', depicted as a bold green box with bold black text. TOGNet processes the combined input to generate a 'Grasp Prediction', shown as a yellow bounding box with a bold black text. The TOGNet also receives input from a 'Target-Oriented Grasp Network (TOGNet)' module, which is a bold green box with bold black text. The final output shows a robotic arm successfully grasping the yellow bottle, visually confirming the system's functionality. The figure effectively demonstrates how TOGNet integrates diverse human inputs—language, gesture, and click—to guide precise robotic grasping, supporting applications in human-robot interaction (HRI) and assistive robotics.

